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Stringent immigration enforcement and the farm sector: Evidence from E-Verify adoption across states

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Abstract

This study examines the impact of E-Verify adoption on the total number of farm workers, total acres of operated farmland, and farm wages. This article finds that states with a high E-Verify enforcement level experiences a decrease in the total number of farm workers and the total acres of operated farmland, suggesting that farms reduced their production scales after E-Verify. However, this study finds no E-Verify impact on farm weekly wages. This result could be a result of crop mix shift. States that adopt more stringent immigration policies should prepare the farm sector for a variety of negative shocks.

KEYWORDS

agricultural workers, E-Verify, farm production, immigration enforcement, labor supply

JEL CLASSIFICATION

J15, J18, J43, O33

E-Verify was launched by the Illegal Immigration Reform and Immigrant Responsibility Act (IIRIRA) in 1996 as a voluntary participation program that compares employees' data to the government records to verify if an employee can lawfully work in the United States. Since the mid-2000s, as a response to the lack of progress on national immigration reform, some states have mandated the use of E-Verify to tackle the issue of illegal immigration by themselves. An effort by the Trump administration for a nationwide mandatory adoption of the E-Verify system was rejected by House leaders in 2018 in part due to strong opposition by farm groups (Irby &

Cadei, 2018). Many US farmers fear they may lose workers and experience an increase in labor costs in the wake of E-Verify adoption (Duehren, 2018). On December 11, 2019, the US House of Representatives passed the Farm Workforce Modernization Act (FWMA) of 2019, which requires mandatory use of E-Verify in the US farm sector. Farm groups are concerned that if the bill becomes law, without other corresponding measures such as guest farm workers reform or undocumented labor legalization, they may suffer economic loss and decrease profitability.

The farm sector is particularly vulnerable to a stringent E-Verify mandate for several reasons. First, in the United States, over 50% of hired farm workers are estimated to be undocumented (Martin, 2016). Second, farm sector jobs are not attractive to domestic workers who are citizens and green card holders due to the high physical demand and low wage and fringe benefits level (Martin & Calvin, 2010; Matthews, 2013). Moreover, the domestic labor force may not fill the void left by undocumented labor if wage levels stay the same. Third, the long and seasonal production cycle, as well as the high proportion of fixed inputs (such as land), lead to a lack of flexibility for the industry to cope with a policy-induced labor decline.

Given that the agenda of a nationwide E-Verify mandate has not been entirely abandoned, and more states are considering E-Verify, this study aims to provide a better understanding of the impact of E-Verify. Using the Synthetic Control Method (SCM) with data from the Quarterly Census of Employment and Wages (QCEW), this study examines the total number of farm workers, total acres of operated farmland, and weekly farm wage in each adopting state. There is a general lack of empirical studies on the impact of tighter immigration policies on agricultural market outcomes (Richards, 2018). Some of the existing research examines the impact of immigration enforcement on the agricultural sector using simulation techniques (e.g., Devadoss & Luckstead, 2011; Zahniser et al., 2012), while several other studies examine the impact of immigration policies on farm labor but look at laws such as the Immigration and Nationality Act Section 287(g) (Charlton & Kostandini, 2021; Kostandini et al., 2014), Immigration Reform and Control Act (IRCA), or the termination of the Bracero program[1] in the 1960s (Clemens et al., 2018). However, limited attention has been given to the impact of E-Verify on the farm sector. To the best of our knowledge, recent relevant studies that examine the impact of E-Verify in the agricultural sector are by Luo et al. (2018), and Luo and Kostandini (2021) and both studies focus on Arizona. More specifically, Luo et al. (2018) find that E-Verify has increased the probability of farm family laborers in Arizona to choose agricultural occupations due to a labor supply shortage. Luo and Kostandini (2021) find a decrease in the labor supply, no wage change, a reduction in labor-intensive crop production, and a decline in the agricultural sector in Arizona. The agricultural sector does not increase the use of farm capital and experiences a decrease in profits.

However, those studies only explore the impact of E-Verify mandate in Arizona, while other enacting states are not examined. In addition, given that the agricultural sector of each of the eight states that have adopted E-Verify laws is quite different in its crop composition, a state-by-state examination sheds light on how the impact differs across states and may provide additional insights to policy makers considering E-Verify mandates.

Since E-Verify implementation levels differ across states (Amuedo-Dorantes & Bansak, 2012), this study examines the effect of E-Verify mandates on the farm sector in each adopting state and how the number of agricultural workers, wages, crop mix, as well as total acres of operated farmland are affected in each state. Some states such as Arizona require all employers to check the eligibility of all newly hired employees. Other states only enforced E-Verify in their public sectors (and contractors) when first passed and extended the law to the private sector later (Amuedo-Dorantes & Bansak, 2012). Some states have exempted the E-Verify mandate for short-term (3-months) employment, such as seasonal farm jobs, and have also exempted firms with few employees (Amuedo-Dorantes & Bansak, 2012). Given these different enforcement levels, and that hired farm labor in the United States varies greatly by geography, size of farms, and crop mix (Martin, 2013), examining the E-Verify impact on a state-by-state basis provides more insightful analysis. The Synthetic Control Method, which mainly focuses on the policy impact of a single treatment unit (a E-Verify adopting state in our case), is suitable for our state-by-state study of E-Verify adoption.

This study has several key findings. First, E-Verify mandates reduced the total number of farm workers in Arizona, Alabama, South Carolina, and Mississippi, which enforced the strictest E-Verify mandates without exemption or mandating a policy phase-in period. The enforcement level of E-Verify in different adopting states plays an important role in determining whether E-Verify would have a significant impact on the farm sector. Second, the decline of farm workers was accompanied by a moderate decrease in the total acres of operated farmland, suggesting that the farm sector reduced production in the wake of E-Verify adoption. Third, we find that the weekly wage rate in the farm sector remained unaltered after E-Verify mandates across all states suggesting a crop mix adjustment. The findings are consistent with the Rybczynski theorem[2] that output adjustment could reduce the pressure on wages caused by labor supply shock.

The remainder of the article is organized as follows: "Background and Conceptual Framework" section introduces the E-Verify adoption across states as well as the farm labor market. "Data And Empirical Strategy" section presents the data and empirical methods. "Empirical Results" section discusses the results. The last section concludes.

BACKGROUND AND CONCEPTUAL FRAMEWORK

E-Verify in the United States

The E-Verify system is a free online program that can be used by employers to verify the employment eligibility of newly hired employees. E-Verify was first launched as a voluntary pilot program by the Illegal Immigration Reform and Immigrant Responsibility Act of 1996. The first E-Verify mandate law was adopted in Arizona in 2008 known as the Legal Arizona Workers Act (LAWA). Bohn et al. (2014) state that LAWA is arguably one of the strictest of the state laws by then because it requires all employers in both public and private sectors to verify the employment eligibility of newly hired workers using the online E-Verify system. The law also imposes employer sanctions, which could result in a business license suspension and revocation for violators. After 2008 comprehensive E-Verify mandates were in Arizona, Alabama, Georgia, North Carolina, Mississippi, South Carolina, Mississippi, Utah, and Tennessee. State governments can choose to adopt E-Verify as well as decide on the specific terms of the mandates. For instance, some states make exemptions for small-scale firms (e.g., Georgia, North Carolina, and Utah) and some choose to phase in the adoption of E-Verify (e.g., Mississippi and North Carolina). Table 1 shows the date of adoption and the differences in E-Verify mandate laws in each state.

Some states grant exemptions to small-scale firms that have 10 or fewer employees (e.g., Georgia) and 14 or fewer employees (e.g., Utah). The exemption rules apply to small farms that hire employees that qualify for the exemption standard (Feere, 2012). In addition, seasonal workers are usually not counted as full-time employees. Thus, farms would report no hired

TABLE 1 Statewide adoption of E-Verify across states

State	Statewide adoption date	Exemptions	Phase-in period
Arizona	Jan. 2008	No	No
Mississippi	July 2008	No	Yes (2008–2011)
Alabama	April 2012	No	No
South Carolina[a]	January 2012	No	No
Utah[b]	July 2010	Yes (14 or fewer employees)	No
Georgia	Jan. 2012	Yes (10 or fewer employees)	Yes (2012–2013)
North Carolina	Oct. 2012	Yes (24 or fewer employees)	Yes (2011–2013)
Tennessee	January 2017	Yes (50 or fewer employees)	No

[a]South Carolina first exempted clergy, the agriculture business and housekeepers employed in private homes from E-Verify program. But amendments to the "South Carolina Illegal Immigration and Reform Act" were signed into law by Governor Nikki Haley on June 27, 2011. The amended law requires all employers to enroll in E-Verify system within three business days.
[b]The E-Verify law in Utah exempts employers of aliens on H-2A (temporary agricultural).
Source: National Conference of State Legislatures (2019).

employees while having several seasonal workers in the field. Some states use phase-in strategies in E-Verify adoption rather than impose a comprehensive implementation at once. For instance, North Carolina required that employers with 500 or more employees use E-Verify effective on October 1, 2012 and then narrowed the requirement to employers with 100 or more employees on January 1, 2013, and finally mandated all employers with 25 or more employees to verify new hires through E-Verify. The phase-in strategy gave local firms some time to adjust their labor demand and granted farm workers some time to change their labor supply (e.g., become self-employed).

Using statistical data from the US Department of Homeland Security, Orrenius et al. (2020) show that in general the use of E-Verify surges after the mandates. There have been concerns that some undocumented works can evade E-Verify by fraudulently utilizing Social Security Numbers (e.g., the case of Gonzalez Hernandez vs. Hernandez Gonzalez in 2021). However, Bohn et al. (2014) show that E-Verify effectively reduced the number of undocumented immigrants in the state of Arizona. In addition, estimates show that 2 years after adoption more than 50% of new hires in Arizona were checked through E-Verify (Bohn et al., 2015). Similarly, 60% of businesses in Alabama and more than 94% of businesses in Georgia were compliant with E-Verify (Ross, 2018). Overall, 970,000 business in the United States have enrolled in E-Verify with 1500 business added weekly (E-Verify.gov, 2021). The E-Verify system shows that despite mandates, there are a number of new hires that do not go through E-Verify. This may undermine the effectiveness of E-Verify and affect our estimates.

Currently, there is a very important piece of legislation, the Farm Workforce Modernization Act of 2021, H.R.1603, which was passed by the House on March 18, 2021, and will benefit the agricultural workforce if also passed by the Senate.[3] This bipartisan immigration legislation would provide undocumented farmworkers and their families with a path to legal status and citizenship. In addition, it would revise the current H-2A agricultural guestworker program and also include new modifications such as increased visa flexibility, housing supply, heat stress protection, cover H-2A workers by the Migrant and Seasonal Agricultural Protection Act, and so forth.

Conceptual framework

This study uses a conceptual framework based on the theoretical model used by Ottaviano and Peri (2012) and Borjas (2014) to predict the wage and crop mix changes in the farm sector in the wake of E-Verify mandates. The conceptual framework has a production function with a nested constant elasticity of substitution between different labor groups (citizen and noncitizen workers[4]) to predict the wage change outcome in the Appendix S1.

E-Verify mandates affect the demand and supply of undocumented workers in the farm sector. As the demand for undocumented workers goes down due to the hiring restrictions imposed by E-Verify mandates, the labor demand of citizen workers would go up and as a result, increase the wage level of citizen workers. On the other hand, as the job opportunities decrease in the area, undocumented workers reduce their labor supply by exiting the farm sector. However, the labor supply of citizen workers may not increase because (1) the changes in the wage and other benefit levels in the farm sector are not substantial enough that to attract more citizen workers to enter the sector; (2) citizen workers are not willing to work in the farm sector due to the physically demanding tasks and seasonality feature of agricultural work. Given the changes in the labor supply and demand of citizen and undocumented farmworkers, it seems that the changes in wage levels are affected by the substitutability between those two groups.

The conceptual framework in the Appendix S1 shows that the wage differential between citizen and noncitizen farmworkers is determined by the elasticity of substitution between those two groups. The degree of substitutability between citizen and noncitizen farmworkers determines the magnitude of the wage impact of E-Verify on the two groups. If citizen and noncitizen workers are perfect substitutes and compete for the same jobs, immigration policies such as E-Verify could have a positive effect on the wage of citizen farmworkers. If citizen and noncitizen workers are imperfect substitutes, the impacts of E-Verify could be insignificant (Card, 2009; Ottaviano & Peri, 2012). On the other hand, the wage level of noncitizen farmworkers who are closer substitutes to undocumented workers could increase (Card, 2009; Ottaviano & Peri, 2012). However, the adoption of E-Verify may enhance the monopsony power of farm employers who could reduce the wage offered to their employees and offset the increasing wage impact of E-Verify. As a result, the wages of citizen and noncitizen farmworkers could remain unchanged in the wake of E-Verify mandates.

Prior research has shown that the farm sector responded to labor shocks caused by the termination of the Bracero program in the 1960s with crop mix change and mechanical power adoption. Those adjustments lower the pressure that could increase wages (Clemens et al., 2018). Gonzalez and Ortega (2011) show that farms can quickly change the crop mix and technology instead of increasing salary level. Gonzalez and Ortega (2011) argue that the adjustment of the crops corresponds to the well-known "Rybczynski effects" in a standard small open economy model. Clemens et al. (2018) also suggest that the Rybczynski type adjustment in the farm sector could explain the result of no substantial wage effect in the sector. This study also tries to find Rybczynski-like effects and predicts that if the crop combination in the farm sector changes from labor-intensive crops (such as specialty crops) to capital-intensive crops (such as grain and feed crops), wage levels in the sector may not be affected.

The farm sector could also use more machines and agricultural technologies to compensate for the loss of labor (Lewis, 2011). The wage level may be restored due to the use of efficient and advanced technologies (Dustmann & Glitz, 2015; Lewis, 2011). Increased adoption of farm mechanization after a labor reduction has been documented by prior studies (Clemens et al., 2018; Lafortune et al., 2015; Richards, 2018). Finally, the total farm output could decrease after labor shock, which would reduce the labor needs of agriculture and therefore the pressure on farm wages (Zahniser et al., 2012).

DATA AND EMPIRICAL STRATEGY

Our analysis uses annual data from the Quarterly Census of Employment and Wages from 2001 to 2018. The QCEW provides a comprehensive employment and wage data set for US workers. It collects employment and wage data from all employers covered by state unemployment insurance programs. All farm employers who employed 10 or more workers in each of 20 weeks of a quarter, or who paid $20,000 or more in quarterly wages in the current or previous year must enroll in the unemployment insurance system and are included in the QCEW. States such as California and Washington require all farm employers to enroll in the unemployment insurance system. In addition, Martin (2020) points out that QCEW estimates the total average farm employment to be about 1.5 million, and the dataset excludes small farm employees and H-2A workers in some states. However, there is no reason to suspect that QCEW treats E-Verify adopting states differently than other states. Additionally, one study by Henry (2015) focused on the impact of E-Verify on the number of H-2A requested by employers and found very little statistical evidence that E-Verify laws across states would lead farms to increasingly rely on the guest farm workers system. Surprisingly, the study found statistical evidence that the strictest E-Verify laws may reduce H-2A usage. The Bureau of Labor Statistics indicates that 92% of wage and salaried agricultural workers are included in the QCEW (Martin, 2013). The use of QCEW data in exploring the farm labor market is also common in previous studies (e.g., Dunn & Hueth, 2017; Hertz & Zahniser, 2013; Martin, 2013; Taylor & Thilmany, 1993). The key dependent variables obtained from the QCEW data are the total number of farm workers (logarithm form) and CPI-deflated weekly farm wages (logarithm form); these two variables are measured for each year at the state level.

Data sources of other variables

The Farm Income and Wealth Statistics of the United States Department of Agriculture Economic Research Service (USDA-ERS) offers information on farm financial indicators such as the total value of production expenses, the value of crop production, and the value of government direct payments to various farm programs. Government direct payments include Price Loss Coverage (PLC), conservation program, supplemental and ad hoc disaster assistance, and Agriculture Risk Coverage (ARC). Total acres of operated farmland (logarithm form) are obtained from the National Agricultura Statistical Service (NASS).[5] In addition, because labor-intensive crops usually have a higher demand for farm labor compared to capital-intensive crops, we include in our analysis the planted acres of vegetables and of field crops. Information on the average farm size is also obtained from the Census of Agriculture. In addition, our regressions control for commodity price of vegetables because it could affect the crop mix and therefore the farm labor hiring decisions in a state. The commodity price of vegetables is calculated by dividing the total production value of vegetables by the total quantity of vegetables measured in hundredweight (cwt). The data for total vegetable production value and quantity are from the USDA National Agricultural Statistics Service.

Data for demographic covariates used in our analysis come from the Current Population Survey (CPS) data. Based on previous studies that examine the labor supply decisions of individual farm workers (Jensen & Salant, 1985; Mishra & Goodwin, 1997; Skoufias, 1994; Sumner, 1982), we include in our analysis the share of noncitizen immigrants among the total population, the share of males among the farm worker population, the share of Hispanics among the population, and the mean age of farm worker population in each state.

Finally, we include a set of real business cycle controls obtained from the Bureau of Economic Analysis (BEA) in our model such as the state-level unemployment rate, per capita government expenditures, per capita personal income, and per capita GDP. Per capita personal income and per capita GDP are deflated using Consumer Price Index, obtained from the Bureau of Labor Statistics. Due to the impact of the 2008 Great Recession, we include the construction sector unemployment rate from the Bureau of Labor Statistics, which calculates the unemployment rate by industry and state based on the CPS. Moreover, Kostandini et al. (2014) and Ifft and Jodlowski (2016) find that other immigration laws, especially 287(g) agreements signed

TABLE 2 Summary statistics of dependent and independent variables

	E-Verify adopting states		E-Verify non-adopting states	
	Mean (1)	Std. dev. (2)	Mean (3)	Std. dev. (4)
Panel A. Dependent variables				
Log total farm workers	9.537	0.612	9.388	1.123
Log total acres of operated farmland	16.15	0.433	15.88	1.701
Log farm weekly wage	6.280	0.180	6.316	0.205
Panel B. Independent variables				
Share of the Hispanic population	0.083	0.093	0.096	0.102
Share of the noncitizen population	0.040	0.027	0.045	0.034
Share of the male population	0.454	0.111	0.460	0.112
Mean age	35.534	9.049	35.148	8.981
Total production expenses	3976.1	2842.8	6043.2	6927.8
Vegetable planted acres	29.0	44.6	27.9	108.4
Field crop planted acres	2606.6	1793.9	6927.9	7880.4
Total crop value	1526.5	1101.7	3439.7	5199.0
Farm size	414.6	572.5	503.3	660.5
Government direct payment	173.5	213.0	218.3	326.7
Real per capita government expenses	5.0	1.9	5.9	2.5
Real per capita personal income	29.7	11.5	35.6	15.0
Real per capita GDP	37.0	13.9	44.3	17.7
Unemployment rate	5.860	2.903	5.069	2.581
Construction unemployment rate	6.311	2.136	5.500	1.960
N	144		684	

with US Immigration and Customs Enforcement (ICE), may affect the farm sector. As a result, we also include controls for 287(g) agreements in our model as well.

Table 2 shows the differences between E-Verify adopting and non-adopting states. E-Verify adopting states hire more farm workers and have more acres of operated farmland in agriculture but have a lower wage level compared to nonadopting states. The farm sector in adopting states focuses more on vegetable production, which leads to significantly lower production of field crops in these states and a lower total value of production and smaller farm size. On the other hand, the demographic characteristics of the farm sector are similar to each other in adopting and nonadopting states. Table 2 shows that E-Verify adopting states, in general, have lower personal income and higher unemployment rates. The higher level of vulnerability in the economy of E-Verify adopting states further raises concerns on the possible adverse impacts of E-Verify on the farm sector in these states.

Empirical strategy: Synthetic control method

We apply the Synthetic Control Method (SCM) to examine the impact of E-Verify adoption on agricultural outcomes at the state level. SCM aims to create well-matched pretrends between treated and control groups by choosing a weighted combination of control states that would create a synthetic treated unit to approximate the relevant characteristics of the real treated unit. This study regards the eight E-Verify adopting states that are presented in Table 1 as treated states and includes the rest of the states in the United States in the control state pool. States that do not have statewide E-Verify mandates in the control state pool are candidates for constructing synthetic treated states. When one E-Verify state such as Arizona is examined by SCM, other E-Verify adopting states are taken away from the sample. The year of implementation is identified if E-Verify mandates become effective in the first 6 months of that year, otherwise the year of implementation is assigned to the next year. As explained below, SCM selects the closest control states to construct the synthetic units for treated units by assigning positive weights. Thus, states in the control group may vary for each treated state. The states with zero weight are not used and the list of states selected with positive weights are available upon request. We include states, which adopted public sector E-Verify in the control group.

In the SCM, the treated state is coded as $j=0$, and the states included in the synthetic control group are coded as $j=(1,2,...,J)$. $t=1,...,T$ denotes the whole sample period and $(1,...,T_0)$ and $(T_0+1,...,\mathrm{T})$ represent pretreatment and posttreatment periods, respectively. The relationship between treated and untreated states j can be presented as follows:

$$Y_{jt}=Y_{jt}^{u}+\eta_{jt}D_{jt}, \tag{1}$$

Y_{jt} denotes the outcome variables focused by this study such as the total number of farm workers in a state. Y_{jt}^{u} represents the outcome variables in untreated states. D_{jt} is a dichotomous variable with a value of 1 for treated state ($j=0$) and years after the treatment ($t>T_0+1$), and a value of 0 otherwise. η_{jt} then estimates the impact of treatment, and we are interested in obtaining η_{jt} when $j=0$ and $t\geq T_0+1$. As a result, we have $\eta_{0t}=Y_{0t}-Y_{0t}^{u}$ where Y_{0t} denotes the actual outcomes of treated state, but Y_{0t}^{u} is unobservable and could be estimated by the model suggested by Abadie et al. (2010) constructed as follows:

$$Y^u_{jt} = \alpha_t + \theta_t Z_j + \mu_j \tau_t + \varepsilon_{jt}, \tag{2}$$

Z_j denotes an $(r \times 1)$ observed predictor vector and θ_t are $(1 \times r)$ estimated coefficients vector. μ_j is a $(k \times 1)$ vector of state specific characteristics that cannot be observed and τ_t is a $(1 \times k)$ vector of unobservable time-varying factors. ε_{jt} denotes the error term. The interaction term $\mu_j \tau_t$ distinguish SCM from conventional difference-in-differences (DID) model because it allows the impacts of confounding unobservable factors specific to each state to vary with time whereas conventional DID does not allow the impacts to vary across time.

Given the model illustrated above, SCM introduces a vector $(J \times 1)$ of weights $W = (w_1, w_2, ..., w_J)$, which are added to 1 and restricted to nonnegative values. Combining those weights with Equation (5) gives the outcome variable for synthetic control unit which can be presented as follows:

$$\sum_{j=1}^{J} w_j Y_{jt} = \alpha_t + \theta_t \sum_{j=1}^{J} w_j Z_j + \tau_t \sum_{j=1}^{J} w_j \mu_j + \sum_{j=1}^{J} w_j \varepsilon_{jt}. \tag{3}$$

There exists an optimal weight vector $W^* = \left(w_1^*, w_2^*, ..., w_J^*\right)$, which helps to construct $Y^u_{0t} = \sum_{j=1}^{J} w_j^* Y_{jt}$ and $Z_0 = \sum_{j=1}^{J} w_j^* Z_j$ for all years within the pretreatment period. With the optimal weights, Abadie et al. (2010) show that the treatment effect of an intervention or a policy for posttreatment period $(T_0 + 1, ..., T)$ can be obtained by $\widehat{\eta}_{0t} = Y_{0t} - \sum_{j=1}^{J} w_j^* Y_{jt}$. The later term is a weighted average of the outcomes from untreated states in control group in the posttreatment period.

To account for such specific trend in Arizona, SCM will match the trend of farmland in Arizona to other states such as Texas, California, and New Mexico that also experienced a decrease in farmland after 2008. Those comparison states would therefore be assigned a higher weight in constructing synthetic Arizona. In addition to the changes in farmland, agricultural production is also different across states. NASS shows that vegetable production in the United States is concentrated in several E-Verify adopting states such as Georgia, South Carolina, North Carolina, and Arizona and in other nonadopting states such as Florida, California, Washington, Wisconsin, and Michigan. Fruit and vegetable production in the United States has been decreasing because the fruit and vegetable imports from Mexico and other countries has been growing for years. By selecting nonadopting states that also experienced a decrease in vegetable and fruit production over the years and assigning more weights to those states, the SCM approach employs a data-driven process to create synthetic units that have the most closely matched pretrends of outcomes to those of the actual E-Verify states while addressing the possible impacts of different pretrends of vegetable and fruit productions across the states in the control group.

To better show the impact of E-Verify mandates, we present the results of aggregated treatment effects by short-run (immediate first year after the enactment) and long-run (the rest years after the enactment). The constructed synthetic treated states provide the counterfactual outcomes to be compared to the actual outcomes of treated states before and after the enactment of E-Verify mandates and the model is presented as follows:

$$\Delta^{\text{shortrun}}_{\text{trt\_state}} = \left| \overline{Y}^{\text{post1yr}}_{\text{trt}_{\text{state}}} - \overline{Y}^{\text{post1yr}}_{\text{syn}_{\text{state}}} \right| - \left| \overline{Y}^{\text{preyrs}}_{\text{trt}_{\text{state}}} - \overline{Y}^{\text{preyrs}}_{\text{syn}_{\text{state}}} \right|, \tag{4}$$

and

$$\Delta^{\text{longrun}}_{\text{trt\_state}} = \left| \overline{Y}^{\text{post1+yr}}_{\text{trt}_{\text{state}}} - \overline{Y}^{\text{post1+yr}}_{\text{syn}_{\text{state}}} \right| - \left| \overline{Y}^{\text{preyrs}}_{\text{trt}_{\text{state}}} - \overline{Y}^{\text{preyrs}}_{\text{syn}_{\text{state}}} \right|, \tag{5}$$

$\overline{Y}^{post1yr}_{trt_{state}}$ and $\overline{Y}^{preyrs}_{trt_{state}}$ in Equation (4) indicate the averaged outcomes of E-Verify mandates enacted states during the posttreatment (immediate first year after enactment) and pretreatment periods, respectively; $\overline{Y}^{post1yr}_{syn_{state}}$ and $\overline{Y}^{preyrs}_{syn_{state}}$ show the averaged outcomes of synthetic states during the posttreatment (immediate first year after enactment) and pretreatment periods, respectively. The variables in Equation (5) are similar but show long-run effects. $\Delta^{shortrun}_{trt\_state}$ and $\Delta^{longrun}_{trt\_state}$ are the changed outcomes experienced by E-Verify mandate states, presenting the impacts of E-Verify in the short and long run.

This study follows the inference method proposed by Abadie et al. (2010) to assess the significance level of SCM estimates. They provide a permutation test, which iteratively treats non-adopting states (placebo states) as the assumed treated states and run SCM for each placebo state. The treated states are omitted from the estimating sample when permutation tests are run. As a result, we construct a distribution of placebo impacts for states that do not adopt E-Verify. As suggested by Abadie et al. (2010), there is a statistically significant impact from E-Verify if the permutation test results show that the estimates of actual adopting states are larger than the majority of the placebo estimates. The formula for calculating the *p*-value of SCM estimates is as follows:

$$p\_\text{value}_t = \text{prob}\left(\widehat{\sigma}_{0t} \leq \widehat{\sigma}^{\text{placebo}}_{jt}\right) = \frac{\sum_{j=1}^{J} S\left(\widehat{\sigma}_{0t} \leq \widehat{\sigma}^{\text{placebo}}_{jt}\right)}{J}, \tag{6}$$

where $S(\cdot)$ is an indicator/switch function, $\widehat{\sigma}_{0t}$ is the actual estimated difference between the outcomes of treated and synthetic unit, and $\widehat{\sigma}^{\text{placebo}}_{jt}$ is the outcome difference between placebo treated and synthetic unit. We exclude placebo states that do not have a well-matched pretrend with their synthetic units in *p*-value calculation.

EMPIRICAL RESULTS

E-Verify impact on the total number of farm workers

Figure 1 presents permutation graphs on the differences in the log total number of farm workers between treated and synthetic states for each year before and after E-Verify adoption. Four states (Alabama, Arizona, South Carolina, and Mississippi) in the top four plots show a decrease in the total number of farm workers in the wake of E-Verify adoption. Moreover, declines in the number of farm workers of those four states (black lines) also differentiate from the trends of other states (gray lines), which indicates a significant decline of farm workers in these states. The four bottom plots (Georgia, North Carolina, Tennessee, and Utah) show no evidence of decreased farm workers and cannot distinguish themselves from other placebo state trends.

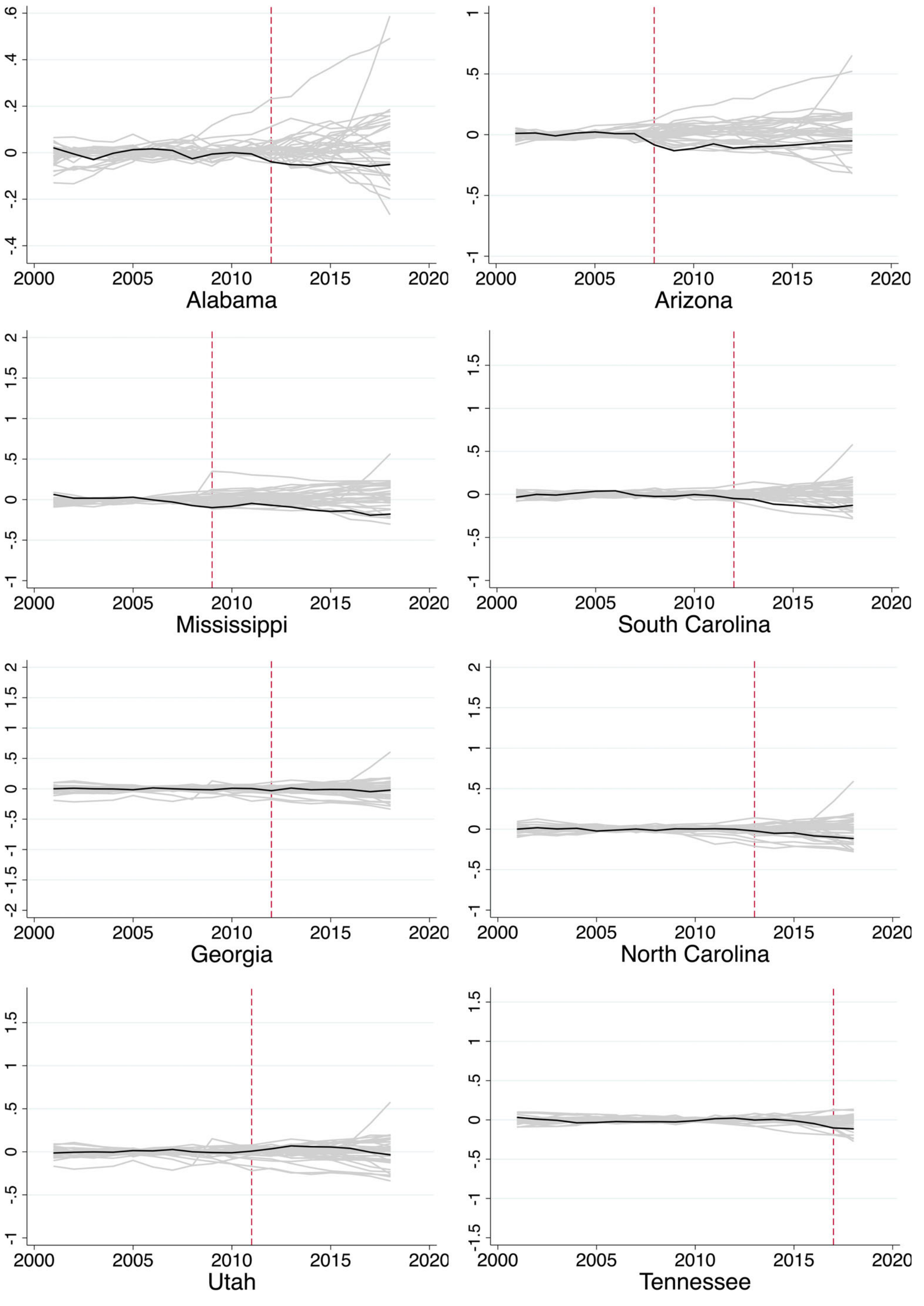

FIGURE 1 Permutation graphs for total number of farm workers differences between treated states and synthetic states

TABLE 3 The synthetic control method estimates of E-Verify impact on the number of farm workers and acres of operated farmland

				DID between 1 year post and pre years		DID between >1 year post and pre years	
	Average diff. between treated and synth. units pretreat years (1)	Average diff. between treated and synth. units 1 year post (2)	Average diff. between treated and synth. units >1 year post (3)	Change post-pre (4)	*p*-value from one-tailed test, P(Δ < ΔAZ) (5)	Change post-pre (6)	*p*-value from one-tailed test, P(Δ < ΔAZ) (7)
Panel A. The number of farm workers							
Alabama	−0.001	−0.039	−0.05	−0.038**	0.023	−0.049	0.14
Arizona	0.01	−0.083	−0.09	−0.093**	0.023	−0.099**	0.023
Mississippi	0.005	−0.099	−0.118	−0.104*	0.051	−0.123	0.102
South Carolina	−0.002	−0.048	−0.133	−0.046**	0.047	−0.132**	0.047
Georgia	0.000	−0.031	−0.022	−0.031	0.419	−0.022	0.465
North Carolina	0.000	−0.022	−0.078	−0.022	0.333	−0.078	0.256
Tennessee	−0.010	−0.103	−0.114	−0.094	0.116	−0.104	0.093
Utah	0.001	0.008	0.031	0.007	0.372	0.031	0.349
Panel B. The acres of operated farmland							
Alabama	−0.002	−0.009	−0.022	−0.007**	0.047	−0.020**	0.047
Arizona	0.000	−0.002	0.004	−0.002**	0.047	0.004	0.333
Mississippi	−0.002	−0.018	−0.023	−0.016	0.310	−0.023	0.153
South Carolina	0.001	−0.002	−0.007	−0.003	0.452	−0.008	0.333
Georgia	0.016	−0.014	0.017	−0.030	0.171	0.001	0.452
North Carolina	−0.014	−0.004	−0.012	0.010	0.358	0.002	0.205
Tennessee	0.001	0.005	0.006	0.004	0.500	0.005	0.571
Utah	0.015	0.014	0.011	0.000	0.119	−0.003	0.571

Note: The average differences in the first three columns are calculated by subtracting the outcome estimates for synthetic Arizona from that for Arizona. The *p*-value estimates are calculated based on the distribution of the estimates for Arizona and other placebo states.

**$p < 0.05$; *$p < 0.1$.

To facilitate the understanding of the graphical evidence shown in Figure 1, we provide the estimates of the averaged differences between treated and synthetic states before and after the adoption and the corresponding *p*-values in Table 3. The differences between treated and synthetic states are averaged by pretreatment period (column 1), first-year posttreatment period (column 2), and more than one-year posttreatment period (column 3). The differences in first-year posttreatment period show the immediate/short-run impact of E-Verify adoption, and the differences in the more than one-year posttreatment period presents the long-run impact on the farm sector outcomes for each state. Columns (4) and (6) show the results by subtracting the differences in column (2) and (3) from that in column (1), respectively. As a result, the estimates in column (4) and (6) are similar to the difference-in-differences estimates showing the overall changes in the farm sector outcomes after the adoption of E-Verify in each adopting state. To evaluate the significance of the outcomes in Table 3, we follow the study of Bohn et al. (2014) and calculate the *p*-value by first constructing a distribution of all E-Verify impact estimates including the treated and placebo states. We then conduct a one-tailed test of the probability of observing a placebo state that has a posttreatment difference in estimate larger than that of treated state. The *p*-values are shown in columns (5) and (7) in Table 3.

Results in column (4) panel A show that Arizona has the largest decrease of 9.3% on the farm worker population in the short run. This is expected as Arizona adopted the first and strictest E-Verify law. The result is consistent with the finding of Luo et al. (2018) that Arizona experienced a decrease in the share of likely undocumented farm workers in the wake of E-Verify adoption. Other states that exhibit a decline in the total number of agricultural workers in the short run have smaller gaps compared to Arizona. Three states exhibit declines in agricultural workers in the long run (column (6) panel A), and they are Arizona, Mississippi, and South Carolina. South Carolina shows the largest decrease in the long run in farm worker population by 13.2%, while Arizona remains at about 9.9% over time. Mississippi shows a decrease in farm workers in the short run by 10.4%, and the estimate is significant at the 10% level. However, a decline of farm workers in Alabama is not evident in column (6), which indicates a fading impact of E-Verify in the state. We do not observe a statistically significant change in Georgia, North Carolina, Utah, and Tennessee in either the short or long run, suggesting that the adoption of E-Verify has no impact in these states.

Four out of eight states experienced a decrease in the total number of agricultural workers. The study by Bohn and Lofstrom (2013) found that the adoption of E-Verify in Arizona has pushed many undocumented workers from wage-and-salary employment into self-employment. The E-Verify mandates also encourage undocumented immigrants to move from formal to informal employment (Gonzalez, 2008). The decreases shown in Figure 1 could be explained by the exit of the undocumented worker from the farm sector or some of them could be employed "off-the-books."

Two plausible reasons can explain the differences in E-Verify impacts across the eight states. First, as shown in Table 1, Arizona, Alabama, Mississippi, and South Carolina have no exemptions for small businesses with few employees. Others for instance Georgia, exempts private employers with less than 10 employees and Utah exempts those with less than 15 employees from using the E-Verify system. This could benefit agricultural employers (Feere, 2012) because many farms in the United States hire a small number of workers. Second, Arizona, Alabama, and South Carolina did not have phase-in periods for E-Verify adoption, while the other states have 2–3 year phase-in periods. The states without a phase-in period face larger shocks in the farm sector since local farms do not have enough time to find a way to retain workers or recruit new ones. As a result, those states experienced a larger and more significant farm labor supply decline after the adoption of E-Verify.

Our findings of decreased numbers of farm workers in four states confirm the concerns of the agricultural industry that a nationwide mandate of E-Verify could exacerbate the labor shortages in the farm sector. We provide empirical and consistent evidence to support the simulation results of Devadoss and Luckstead (2011) and Zahniser et al. (2012) that suggest tightened immigration enforcement could cause labor endowments to decrease in agriculture. We further point out that the decrease of farm labor could be conditioned on the enforcement level of E-Verify mandates. This study suggests that more rigidly enforced E-Verify mandates would be more likely to cause notable labor decreases.

E-Verify impact on acres of operated farmland

Given restrictions in farm technology adoption, particularly in labor-intensive crops, one of the immediate responses of the farm sector to tighter immigration policies would be to reduce the agricultural production in a state to adjust to a smaller labor pool. Figure 2 shows the permutation graphs for the changes in the log acres of operated farmland in each E-Verify adopting state. The total acres of operated farmland represent the total land input in a state each year after E-Verify adoption, and it reflects the scale of agricultural production in each state.

Figure 2 shows that the total acres of operated farmland in Alabama decreases by 0.7% in the first year and 2% over time. The total acres of operated farmland in Arizona shows an instant decrease by 0.2% after the adoption (Table 3). However, the decreasing trend only lasts for 3 years and then becomes statistically insignificant in the long run. South Carolina and Mississippi do not show notable changes in the operated farmland following the adoption of E-Verify and as shown earlier (panel B in Table 3); the estimates of E-Verify impact are not significant either in the short run nor in long run. However, Figure 2 shows that the decreases in the acres of operated farmland in South Carolina and Mississippi begin to stand out from the rest of the placebo states in 2017 and 2018. This suggests a delayed decrease in the scale of farm production. The results of operated acres of farmland in these four states are broadly in line with the findings of a farm working population decline in the wake of E-Verify adoption. Georgia, North Carolina, Utah, and Tennessee do not experience a decrease in farm labor and show no change in their farm production.

Our findings of decreased acres of operated farmland support the observations made by Huffman (2007) who suggests that the farm sector usually responds to labor reduction first by means of production downsizing rather than raising wages higher to attract US farm workers. It is the demand side of the labor input that dominates the responses to a wage increase but not the supply side.

E-Verify impact on weekly farm wage

Results so far indicate that the adoption of E-Verify caused a decline in the number of farm workers in four states. However, farm wages may or may not change because they depend on the adjustment of farm technology use, agricultural production value, and crop mix (Clemens et al., 2018). Figure 3 displays the results of E-Verify mandate impacts on farm weekly wage in each state, and they are consistent with our predictions. Permutation tests in Figure 3 show that

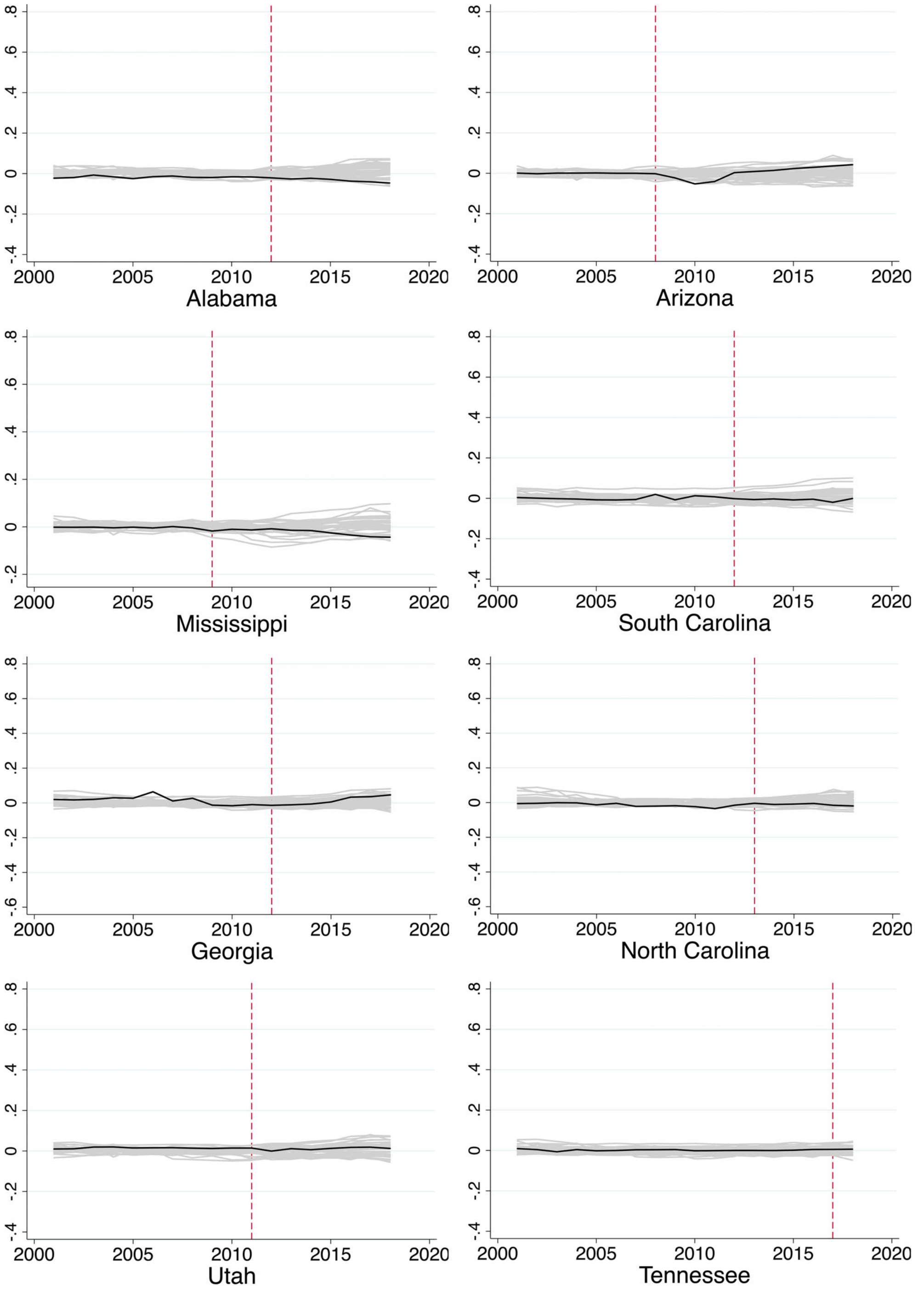

FIGURE 2 Permutation graphs for total acres of agricultural operated land differences between treated states and synthetic states

the farm weekly wages do not exhibit significant changes in all states. The changes in the weekly farm wage in E-Verify adopting states do not stand out from the rest of the states;

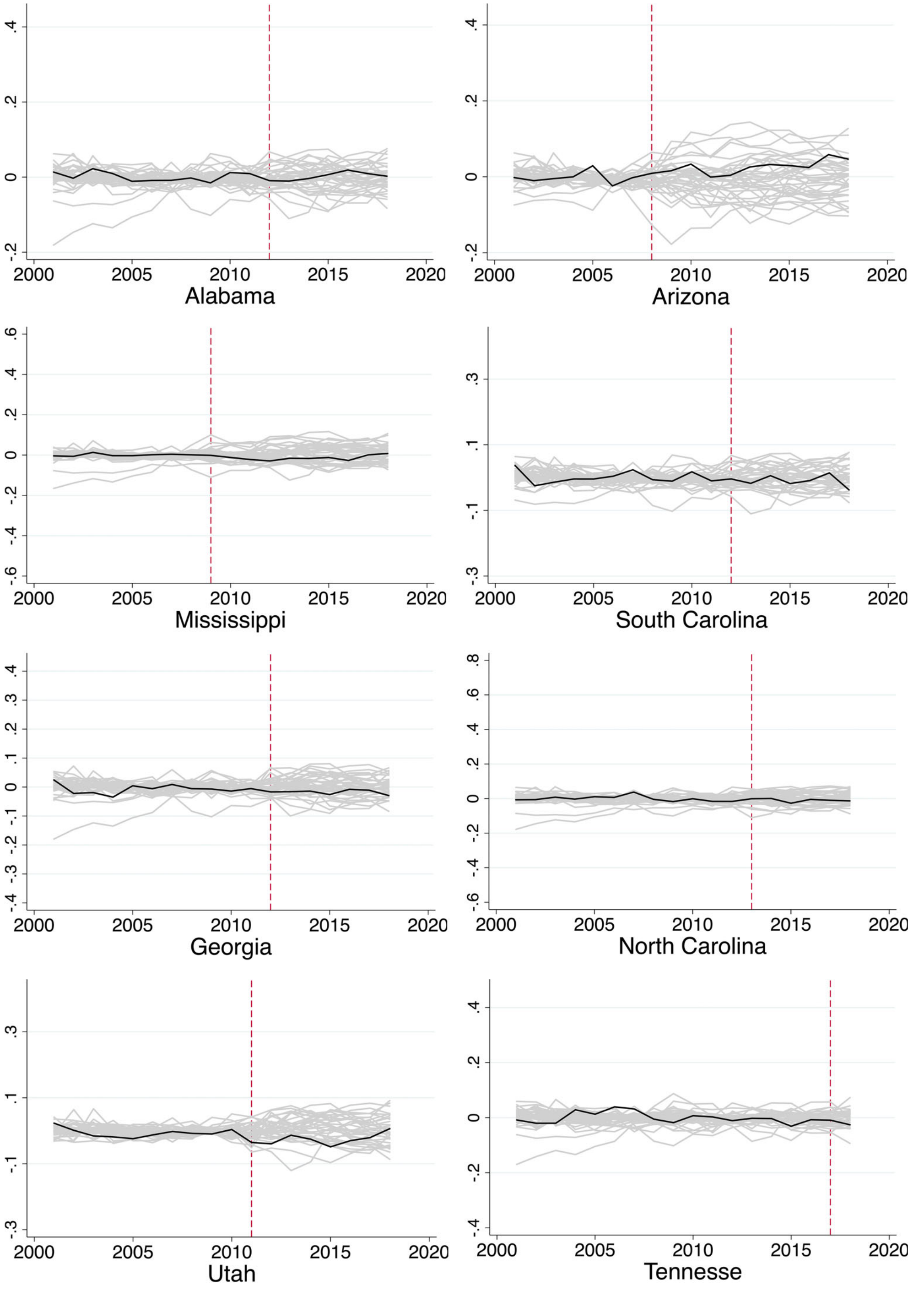

FIGURE 3 Permutation tests for weekly wages differences between treated states and synthetic states

therefore, there is no evidence suggesting that weekly farm wage has experienced a notable change after the adoption of E-Verify.

TABLE 4 The synthetic control method estimates of E-Verify impact on farm weekly wages

				DID between 1 year post and pre years		DID between >1 year post and pre years	
	Average diff. between treated and synth. units pre-treat years (1)	**Average diff. between treated and synth. units 1 year post (2)**	**Average diff. between treated and synth. units >1 year post (3)**	**Change post-pre (4)**	***p*-value from one-tailed test, P ($\Delta < \Delta AZ$) (5)**	**Change post-pre (6)**	***p*-value from one-tailed test, P($\Delta < \Delta AZ$) (7)**
Alabama	0.002	−0.009	0.004	−0.011	0.163	0.002	0.326
Arizona	−0.002	0.009	0.027	0.011	0.256	0.029	0.140
Mississippi	0.000	0.002	−0.014	0.002	0.487	−0.014	0.384
South Carolina	0.001	−0.004	−0.010	−0.005	0.488	−0.011	0.395
Georgia	−0.007	−0.017	−0.017	−0.010	0.347	−0.010	0.327
North Carolina	0.000	−0.001	−0.011	−0.001	0.461	−0.011	0.358
Tennessee	0.000	−0.009	−0.026	−0.009	0.372	−0.025	0.186
Utah	−0.006	−0.035	−0.025	−0.029	0.140	−0.018	0.209

Note: The average differences in the first three columns are calculated by subtracting the outcome estimates for synthetic Arizona from that for Arizona. The *p*-value estimates are calculated based on the distribution of the estimates for Arizona and other placebo states.

Arizona, Alabama, South Carolina, and Mississippi experienced a decrease in the total number of farm workers and a decline in acres of operated agricultural land but no change in farm wages. The insignificant and small farm wage changes shown from columns (4) to (7) of Table 4 provide the estimates of the pre- and posttreatment differences and the corresponding *p*-values and are in line with the visual evidence in Figure 3. The outcomes of weekly wages for farm workers in E-Verify adopting states remain unchanged in both the short run and long run.

E-Verify impact on crop mix and labor productivity

Supporters of stringent immigration enforcement argue that farms can attract more domestic workers by raising farm wages (Richards, 2018). However, that may not be the case in the US agricultural sector since a wage change is subject to factors such as the crop mix and labor productivity that could reflect machine use.

Output mix or technologies are regarded as two major mechanisms that could be employed by the farm sector in the wake of labor supply shock (Lafortune et al., 2015). Table 5 provides the outcomes of log total acres of vegetable crops, log total acres of field crops, and farm labor productivity (the total agricultural production value per worker). The first two indicators are

TABLE 5 The DID estimates of E-Verify impact on the crop mix and farm labor productivity

	Field crop acres (1)	Vegetable crop acres (2)	Labor productivity (3)
Alabama	0.156*** (0.049)	−0.651** (0.280)	47.004** (17.649)
Arizona	0.282*** (0.054)	−0.394** (0.192)	1.667 (24.727)
Mississippi	0.220*** (0.056)	−0.002 (0.157)	8.096 (14.533)
South Carolina	0.04 (0.089)	−0.172 (0.266)	−28.719 (20.388)
Georgia	−0.012 (0.069)	−0.677 (0.476)	73.356*** (25.092)
North Carolina	0.116* (0.064)	−0.691** (0.268)	37.085 (31.527)
Tennessee	−0.370* (0.191)	−0.617 (0.484)	−4.929 (83.532)
Utah	0.021 (0.078)	0.017 (0.264)	−26.793 (23.233)

Note: The covariates included in estimation are the state-level unemployment rate, per capita personal income, the construction sector unemployment rate, the value of government direct payment, vegetable average price, 287(g) agreement adoption status. The covariates for different outcomes may vary. Robust standard errors are clustered at the state level.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

used to investigate the changes in the crop mix in each state and the farm labor productivity is employed to examine if the farm sector adopted any measure (e.g., machines and new technologies) to increase the labor productivity to compensate for the lost labor in the wake of E-Verify adoption. The crop mix data are from the annual NASS survey, and labor productivity is calculated based on the data from the Farm Income and Wealth Statistics. The vegetable production data provided by NASS cover the acres of vegetable grown in each state by year. Due to data limitation, the data series do not include the production acres of other relevant crops such as horticultural crops and fruits. However, our findings of the vegetable production sector could have implications for those sectors because they are also labor-intensive, highly profitable, and facing high pressure from foreign competition. A DID model is used to generate the results in Table 5 for each state, and the model is specified as follows:

$$\text{outcome}_{st} = \alpha + \beta_1 \text{Everify}_{ist} + \beta_2 X'_{st} + \delta_t + \theta_s + \varepsilon_{ist}, \tag{7}$$

where outcome_{ist} denotes the dependent variables indicating the field crop acres, vegetable crop acres, and labor productivity in state s and year t. Everify_{st} equals one if a state adopted statewide E-Verify mandate after the year of adoption and zero otherwise. β_1 measures the impacts of E-Verify adoption on the farm sector's crop mix and labor productivity. X'_{st} is a vector of covariates included in the model. δ_t and θ_s are year-fixed effect and state-fixed effect, respectively. ε_{ist} is the error term. Robust standard errors are clustered at the state level.

The results from column (1) and (2) of Table 5 show that many states, including those that do not show a significant decrease in the number of farm worker (e.g., Georgia and North Carolina), experience a decrease in the total acres of vegetable crops. On the other hand, the acres of field crops that are more mechanized and rely less on the labor input increase in Alabama, Arizona, and North Carolina. The decreases in the total acres of vegetable crops and increases in field crops in many states indicate that the farm sector resorts to crop mix in the wake of labor shock. The notable change in crop mix in the states that does not show a significant change in the total number of farm workers after E-Verify may be due to the reduced farm working time supplied by farm workers.

In addition, we do not see any state that experienced a significant increase in farm labor productivity except Alabama. The increase in the labor productivity found in Alabama could be because the farm sector of the state adopted more machines to aid the staying farm labor in their work or that the farm workers in field crops in general have higher productivity and the shift in crop mix, therefore, resulted in increased the labor productivity. The insignificant change in labor productivity in other states is expected because it reflects the stagnated mechanization situation that is faced by many farms, particularly for specialty crops, in the United States nowadays. As Huffman (2012) points out, little progress in mechanization has been made within the vegetable and fruit production sector in recent decades compared to grain and feed crops. The limited breakthrough in vegetable and fruit production means that the farms cannot adopt more machines to replace labor inputs and therefore can only reduce the production of labor-intensive crops. As labor productivity remains the same, farms lack the motivation to raise the farm wage to attract workers. Hence, there is little wage incentive for domestic workers to enter the farm sector. As a result, the farm sector may experience a decreased production scale, stagnated wage level, loss of crop diversity, and reduced profitability in the wake of E-Verify adoption.

CONCLUSIONS

From 2008 to2018, eight states adopted comprehensive E-Verify mandates that require all employers to verify the employment eligibility of newly hired workers to curb the hiring of undocumented immigrants and improve the labor market outcomes for domestic workers (Orrenius & Zavodny, 2015). Using data from multiple sources, we find that comprehensive E-Verify mandates reduced the total number of farm workers in four adopting states that use the most restrictive E-Verify mandates without exemption and policy phase-in period. Moreover, an investigation of the total acres of operated farmland reveals that the farm sector decreased the scale of production after E-Verify. We find no significant changes in the wages of farm workers due to E-Verify, which indicates that, so far, E-Verify did not improve the farm wage level as it intended.

This article indicates that states with less restrictive E-Verify may give local economies or farms time to adjust to or search for a way to retain farm population; states that did not would experience a decrease in farm labor. Moreover, findings for adopting states that experienced a decrease in farm labor indicate they reduce acres of operated farmland when no farm workers are available for hiring. With the adjustments in farm production and crop mix, a farm wage rise that is expected by strict immigration control advocates becomes an unlikely event. The unchanged farm wage would not help fill the void of decreased farm labor and farms, particularly the specialty crop sector, may lack proper mechanical aids and effective managerial solutions to cope with the tighter immigration enforcement across the country. National and state policymakers and relevant stakeholders need to be more aware of the possible repercussions of imposing E-Verify in the farm sector.

ENDNOTES

[1] The Bracero program was an agreement signed in 1942 between the U.S. and Mexican governments that gave working permits to Mexican citizens to work in the US farm sector.

[2] Rybczynski theorem suggests that in a small open economy such as a state (Hanson & Slaughter, 1999; Hanson & Slaughter, 2002), when the agricultural commodity prices are externally decided, the impact of a change in one of the input factors would be absorbed by production adjustment such as crop mix change. As a result, the wage level in that state's farm sector could be the same.

[3] For additional information visit: https://www.congress.gov/bill/117th-congress/house-bill/1603/text and https://www.farmworkerjustice.org/wp-content/uploads/2021/03/FarmWorkforceModernizationAct-FactSheet-FJ-2021.pdf.

[4] We note that "citizen" and "noncitizen" in the theoretical framework mean US-born and foreign-born individuals, respectively. There is a small share of population who are foreign-born naturalized citizens but it does not change the results of our theoretical framework.

[5] Given a considerable share of vegetable production in the U.S. takes place in greenhouse and protected agriculture, we searched for information on whether this data is fully captured by NASS data for each state. We contacted NASS main offices and several other state field offices and they all mentioned that NASS data does not include greenhouse or protected agriculture production in any state.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

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